

Literature Status for the ℓ_1 NQP-Basis Question

FA/Banach run packet

June 14, 2026

Status

Result type: literature already answered, basis subquestion.

Source: S. J. Dilworth, E. Odell, Th. Schlumprecht, and A. Zsak, *Coefficient Quantization in Banach Spaces*, arXiv:math/0606317.

Supporting source: P. G. Casazza, S. J. Dilworth, E. Odell, Th. Schlumprecht, and A. Zsak, *Coefficient Quantization for Frames in Banach Spaces*, arXiv:0711.2484.

Source Question

In Section 5 of arXiv:math/0606317, PDF page 21, the authors ask:

Does ℓ_1 have an NQP basis (resp. minimal system)?

The arXiv PDF labels this as Problem 5.17.

Supporting Answer

In Section 5 of arXiv:0711.2484, PDF page 30, Corollary 5.15 states that an infinite-dimensional Banach space with nontrivial cotype cannot have a semi-normalized basis with the NQP. The corollary then explicitly says, citing the earlier DOSZ problem, that ℓ_1 does not have a semi-normalized basis with the NQP.

This gives a negative answer to the intended semi-normalized basis version of the ℓ_1 question, because ℓ_1 has nontrivial cotype.

Scope and Limitations

This is a literature-status packet, not a new proof. It records only the semi-normalized basis subquestion. It does not claim the “minimal system” part of the source question, and it does not settle the broader source Problem 5.16 asking whether every Banach space with an NQP basis contains c_0 .

There is a numbering mismatch: the source arXiv PDF labels the question as Problem 5.17, while the supporting paper cites “Problem 5.18 in [DOSZ]”. The cited wording and context identify the same ℓ_1 NQP-basis question.

Search Evidence

The run's cheap indexes were searched for arXiv:math/0606317, arXiv:0711.2484, NQP, coefficient quantization, and the ℓ_1 NQP-basis wording; no prior packet or attempt was found. A phrase search for the exact NQP terminology found the frame follow-up, whose TeX/PDF source was then checked locally.

References

References

- [1] S. J. Dilworth, E. Odell, Th. Schlumprecht, and A. Zsak, *Coefficient Quantization in Banach Spaces*, arXiv:math/0606317. <https://arxiv.org/abs/math/0606317>
- [2] P. G. Casazza, S. J. Dilworth, E. Odell, Th. Schlumprecht, and A. Zsak, *Coefficient Quantization for Frames in Banach Spaces*, arXiv:0711.2484. <https://arxiv.org/abs/0711.2484>